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KPK BOARD TEXTBOOK NOTES - MATHEMATICS CLASS 9

Unit 14: Ratio and Proportion

Complete, easy English notes based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - ratio and proportion, similar triangles, and the four theorems of the unit with full proofs, worked examples and exam practice.

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> **How to use these notes:** This is a theorem unit. First learn each theorem statement word for word, then follow its proof step by step, then try the worked examples. In the board paper the theorems carry long-question marks, and the exercise questions are short applications of the same four results.

What this unit covers

1. Ratio and proportion - meaning, simplest form, and the cross-multiplication test
2. Similar triangles - correspondence and the three similarity tests (AA, SSS, SAS)
3. Theorem 14.1 - a line parallel to one side of a triangle divides the other two sides proportionally, with its corollary
4. Theorem 14.2 - the converse: a line dividing two sides in the same ratio is parallel to the third side
5. Theorem 14.3 - the internal bisector of an angle divides the opposite side in the ratio of the sides containing the angle
6. Theorem 14.4 - in similar triangles, corresponding sides are proportional
7. Applications - unknown lengths, angle ratios, midpoints and areas of similar figures
8. Theorem summary, common mistakes, MCQs and practice questions with answers

Source book: KPTB Mathematics for Grade IX, Unit 14 (book pages 275-284). The notes follow the book's order, theorem numbering and exercise. The scan used for verification carries no printed copyright page, so no printing year is claimed here - the 17-unit structure is the current circulating KPTB edition.

1. Ratio and Proportion

Ratio

A **ratio** is a comparison of two quantities of the **same kind**, measured in the same unit. The ratio of *a* to *b* is written **a : b** or as the fraction **a/b**, where b is not zero. The first term *a* is the **antecedent** and the second term *b* is the **consequent**.

Example 1 (from the book). Area of triangle ABC = 18 cm^2 and area of triangle DEF = 24 cm^2. The ratio of their areas is 18 : 24. Dividing both terms by 6 gives the **simplest form, 3 : 4**. Always reduce a ratio to its simplest form, just like a fraction.

- A ratio has **no unit** - the units cancel, because both quantities are of the same kind.
- Convert to the same unit first: 2 m : 50 cm = 200 : 50 = **4 : 1**, not 2 : 50.
- Multiplying or dividing both terms by the same non-zero number does not change the ratio.

Proportion

When the ratio of *a* to *b* is the same as the ratio of *c* to *d*, the four quantities are in **proportion**. We write **a : b :: c : d**, or **a/b = c/d**. Here *a* and *d* are the **extremes**, and *b* and *c* are the **means**.

Cross-multiplication test: a/b = c/d if and only if **a x d = b x c** (product of extremes = product of means). Use this to test a proportion and to find a missing term.

Example 2. Are 4, 6, 10, 15 in proportion? 4 x 15 = 60 and 6 x 10 = 60. The products are equal, so **yes**: 4 : 6 :: 10 : 15.

Example 3. Find x if 3 : 5 :: x : 20. 5x = 3 x 20 = 60, so **x = 12**.

Two properties the book uses in this unit

- **Componendo:** if a/b = c/d, then (a + b)/b = (c + d)/d - add 1 to both sides.
- **Invertendo:** if a/b = c/d, then b/a = d/c - turn both fractions upside down.

The book applies these on page 276: if in a triangle AD/DB = AE/EC, then by componendo AB/DB = AC/EC, and by componendo with invertendo, AD/AB = AE/AC. You will meet all three forms in Theorem 14.1 and its corollary - learn to move between them.

2. Similar Triangles

Two triangles are **similar** when they have the same shape but not necessarily the same size. In a correspondence triangle ABC <-> triangle DEF, if the corresponding angles are congruent - angle A = angle D, angle B = angle E and angle C = angle F - the triangles are similar, written **triangle ABC ~ triangle DEF**. This correspondence itself is called **similarity** (book, page 276).

> **Order matters:** In ABC ~ DEF the letters are in corresponding order: A matches D, B matches E, C matches F. Most errors in this unit come from matching the wrong sides. Always write the similarity statement in order first, then read the ratios off it.

The three similarity tests (book summary, page 284)

Test	What to check
AA test	Two angles of one triangle are equal to two angles of the other triangle. (The third angles are then automatically equal.)
SSS test	All three pairs of corresponding sides are proportional - the three ratios are equal.
SAS test	One angle of one triangle equals an angle of the other, and the sides that include (form) that equal angle are proportional.

SSA is not a valid test - this exact point is asked in the book's Review Exercise 14. Two sides and a non-included angle do not fix the shape of a triangle.

Congruent is a special case of similar

Congruent triangles are similar with ratio 1 : 1 - same shape *and* same size. Similar triangles need only the same shape: every length in one is the corresponding length in the other multiplied by one fixed number, the **ratio of similarity**.

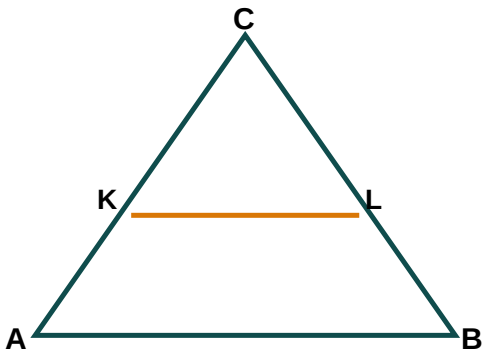
Example 4. Triangle ABC ~ triangle DEF with AB = 4 cm and DE = 8 cm. The ratio of similarity (ABC to DEF) is 4 : 8 = 1 : 2, so *every* side of DEF is double the matching side of ABC, and every angle is unchanged. If BC = 5 cm, then EF = 10 cm.

> **How the unit fits together:** Theorem 14.1 produces equal ratios inside one triangle. Theorem 14.4 is the bridge the other way - once two triangles are known to be similar (by AA, SSS or SAS), all their corresponding sides are in one proportion, and unknown lengths can be found.

3. Theorem 14.1 - A Parallel Line Divides Two Sides Proportionally

Theorem 14.1. A line parallel to one side of a triangle, intersecting the other two sides, divides them proportionally.

Given: In triangle ABC, line KL is parallel to AB. KL meets CA at K and CB at L. **To prove:** CK/KA = CL/LB.



Construction: Take the unit of length so that CK = p units and KA = q units (p, q whole numbers, q not 0), so CK/KA = p/q. Divide CK into p congruent segments and KA into q congruent segments, and from each dividing point draw a line parallel to KL.

Statements	Reasons
The parallel lines divide CA into (p + q) congruent segments.	Construction
CB is another transversal cutting these parallel lines, so they divide CB also into (p + q) congruent segments; CL gets p of them and LB gets q.	Equal intercepts on transversals
If each small segment has length a, then CL/LB = (a x p)/(a x q) = p/q ... (i)	By counting segments
But CK/KA = p/q ... (ii)	Construction
Therefore CK/KA = CL/LB.	From (i) and (ii)

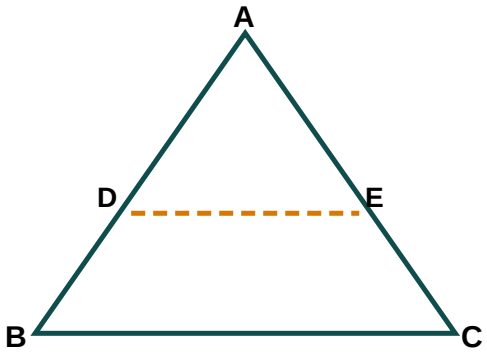
Corollary - the two whole-side forms

In triangle ABC, if KL || AB, then **(i) CA/KA = CB/LB** and **(ii) CK/CA = CL/CB**. Both follow from the theorem by componendo and invertendo (page 2). In the common D-E notation (DE || BC in triangle ABC): **AD/DB = AE/EC**, and also **AD/AB = AE/AC** and **AB/DB = AC/EC**. Use whichever form contains the three known lengths.

4. Theorem 14.2 - The Converse of Theorem 14.1

Theorem 14.2. If a line segment intersects two sides of a triangle and divides them in the same ratio, then it is parallel to the third side.

Given: In triangle ABC, D lies on AB and E lies on AC, and AD/DB = AE/EC. **To prove:** DE || BC.



Proof in four steps

1. Suppose DE is **not** parallel to BC. Through D draw a line parallel to BC, meeting AC at a point F (different from E).
2. By Theorem 14.1 applied to this parallel line, AD/DB = AF/FC.
3. But it is given that AD/DB = AE/EC, so AF/FC = AE/EC. By componendo, AF/AC = AE/AC, which gives **AF = AE**.
4. So F and E are the same point - contradicting the supposition. Therefore **DE || BC**.

> **How to use the pair:** Theorem 14.1 starts from "parallel" and gives equal ratios - use it to *find lengths*. Theorem 14.2 starts from equal ratios and gives "parallel" - use it to *prove a line parallel*. Exam questions that ask "show that the line is parallel" are always Theorem 14.2.

Example 5 (book, Exercise 14.1 Q5). In triangle ABC, BC || DE with AD = 1.5 cm, BD = 3 cm and AE = 1.3 cm. Find CE.

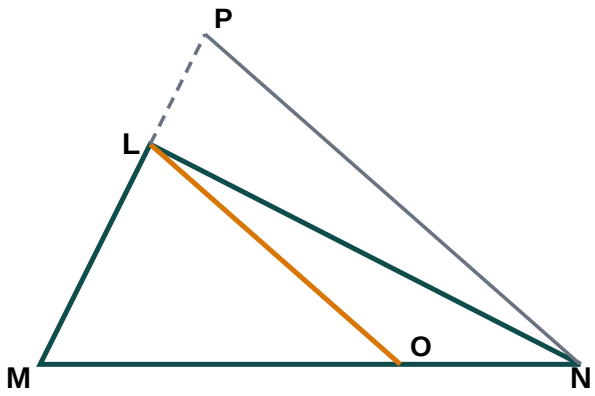
By Theorem 14.1: AD/DB = AE/EC, so 1.5/3 = 1.3/EC. Cross-multiplying: 1.5 x EC = 3.9, so **CE = 2.6 cm**.

Example 6. D on AB and E on AC with AD = 2 cm, DB = 4 cm, AE = 3 cm, EC = 6 cm. Since AD/DB = 2/4 = 1/2 and AE/EC = 3/6 = 1/2, the ratios are equal, so by Theorem 14.2, **DE || BC**.

5. Theorem 14.3 - The Internal Bisector Theorem

Theorem 14.3. The internal bisector of an angle of a triangle divides the side opposite to it in the ratio of the lengths of the sides containing the angle.

Given: In triangle LMN, LO bisects angle L and meets MN at O. **To prove:** MO/NO = LM/LN.



Construction: Through N draw NP || LO, meeting LM produced at P.

Statements	Reasons
Angle MLO = angle OLN ... (i)	LO bisects angle L (given)
Angle OLN = angle PNL ... (ii)	Alternate angles, NP || LO, transversal LN
Angle MLO = angle LPN ... (iii)	Corresponding angles, NP || LO, transversal PM
From (i), (ii), (iii): in triangle PLN, angle LPN = angle PNL, so the sides opposite them are equal: **LN = LP**.	Base angles equal, so triangle PLN is isosceles
In triangle PMN, LO || PN, so by Theorem 14.1: NO/MO = LP/LM. Turning both ratios over and replacing LP by LN: **MO/NO = LM/LN**.	Theorem 14.1 and LN = LP

In the usual A-B-C naming - AD the bisector of angle A meeting BC at D - the result is **BD/DC = AB/AC**: the two parts of BC are in the ratio of the two sides that form angle A.

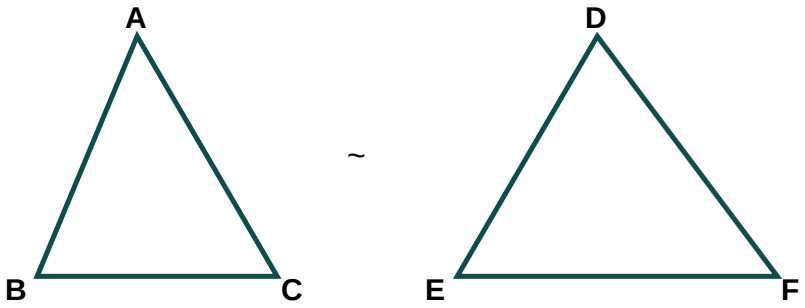
Example 7 (book, Exercise 14.1 Q7). AD bisects angle A of triangle ABC, with AB = 9 cm, AC = 10 cm, BC = 12 cm. Find BD and DC.

BD/DC = AB/AC = 9/10 and BD + DC = 12. So BD = 12 x 9/19 = 108/19 = **5.68 cm** and DC = 12 x 10/19 = 120/19 = **6.32 cm** (check: 5.68 + 6.32 = 12).

6. Theorem 14.4 - Similar Triangles Have Proportional Sides

Theorem 14.4. If two triangles are similar, then the measures of their corresponding sides are proportional.

Given: Triangle ABC ~ triangle DEF in the correspondence ABC <-> DEF. **To prove:** AB/DE = BC/EF = CA/FD.



Proof idea (book, pages 280-281)

1. **Construction:** On DE and DF mark points G and H so that DG is congruent to AC and DH is congruent to AB. Then triangles ABC and HDG are congruent (SAS: two sides and the included angle A = angle D).
2. So angle DHG = angle B = angle E (corresponding angles), which gives **HG || EF**.
3. By Theorem 14.1 in triangle DEF: DH/DE = DG/DF. Replacing DH by AB and DG by AC: **AB/DE = AC/DF** ... (i)
4. Repeating the same argument with the construction made at another vertex gives **AC/DF = BC/EF** ... (ii)
5. From (i) and (ii): **AB/DE = BC/EF = AC/DF**, which is what was to be proved.

Example 8 (book, Exercise 14.1 Q12). Triangle ABC ~ triangle DEF, with AC = 12 cm, DF = 9 cm, AB = 4 cm, BC = 10 cm. Find DE and EF.

AB/DE = BC/EF = AC/DF = 12/9 = 4/3. From AB/DE = 4/3: 4/DE = 4/3, so **DE = 3 cm**. From BC/EF = 4/3: 10/EF = 4/3, so EF = 30/4 = **7.5 cm**.

Example 9 (book, Exercise 14.1 Q11 - checking similarity). Sides 16, x, 28 in one triangle correspond to 22, 33, 38.5 in another. 16/22 = 8/11 and 28/38.5 = 8/11, so the triangles are similar by the SSS test only if x/33 = 8/11 as well, giving **x = 24**.

7. Using the Theorems - Angles, Midpoints and Areas

Dividing a total in a given ratio

Example 10 (book, Exercise 14.1 Q1). The angles of a triangle are in the ratio 1 : 2 : 6. Find the smallest angle.

Total parts = 1 + 2 + 6 = 9, and the angles sum to 180 degrees, so one part = 180/9 = 20 degrees. The smallest angle is **20 degrees** (the others are 40 and 120 degrees).

Example 11 (book, Exercise 14.1 Q2 pattern). Angles in the ratio 2 : 3 : 4: total 9 parts, one part 20 degrees, so the largest angle is 4 x 20 = **80 degrees**. The same "total parts" method splits any quantity - for example 32 students in the boys-to-girls ratio 3 : 5 gives 12 boys and 20 girls, a difference of 8 (book, Review Exercise 14).

The midpoint result

If D and E are the **midpoints** of AB and AC, then AD/DB = AE/EC = 1, so by Theorem 14.2, DE || BC - and triangle ADE ~ triangle ABC with ratio 1 : 2, so **DE = (1/2) BC**. The book's Review Exercise uses exactly this: the segment joining the midpoints measures 3.5 cm, so the third side is **7 cm**.

Areas of similar figures (book summary, page 284)

The ratio of the **areas** of two similar figures equals the **square** of the ratio of their corresponding lengths: Area 1 / Area 2 = (length 1 / length 2)^2.

Example 12. Two similar triangles have corresponding sides in the ratio 2 : 3. Their areas are in the ratio 2^2 : 3^2 = **4 : 9**. If the smaller area is 36 cm^2, the larger is 36 x 9/4 = **81 cm^2**.

Do not mix the ratios: lengths compare directly (2 : 3), perimeters also compare directly (2 : 3), but areas compare as squares (4 : 9). Writing 2 : 3 for areas is a classic lost mark.

8. Theorem Summary and Common Mistakes

Result	Statement in one line
Theorem 14.1	Line parallel to one side divides the other two sides proportionally: AD/DB = AE/EC (for DE || BC).
Corollary	AD/AB = AE/AC and AB/DB = AC/EC - the part-to-whole and whole-to-part forms.
Theorem 14.2	Converse: equal division ratios on two sides mean the line is parallel to the third side.
Theorem 14.3	The internal bisector divides the opposite side in the ratio of the containing sides: BD/DC = AB/AC.
Theorem 14.4	Similar triangles have corresponding sides proportional: AB/DE = BC/EF = CA/FD.
Similarity tests	AA, SSS (sides proportional), SAS (included angle equal, its sides proportional). SSA is not a test.
Areas	Area ratio = square of the corresponding-length ratio.

Five mistakes that lose marks

1. Matching sides from the similarity statement in the wrong order - write ABC ~ DEF in order and read ratios straight from it.
2. Using part-to-part (AD/DB) on one side and part-to-whole (AE/AC) on the other in the same equation.
3. Applying Theorem 14.1 when the line is not stated or proved parallel - prove it first, or use Theorem 14.2 the other way.
4. In the bisector theorem, pairing BD with AC instead of AB - each segment of the base pairs with the side on its own side of the triangle.
5. Giving the area ratio unsquared, or squaring a length ratio by mistake.

> **One-line revision:** Parallel gives equal ratios (14.1), equal ratios give parallel (14.2), the bisector splits the base like the sides (14.3), and similar triangles share one ratio for all sides (14.4).

9. Practice - MCQs and Questions

Multiple-choice questions

1. The ratio 18 : 24 in simplest form is: (a) 6 : 8 (b) 3 : 4 (c) 4 : 3 (d) 9 : 12
2. In triangle ABC, DE || BC, AD = 2 cm, DB = 4 cm, AE = 3 cm. Then EC equals: (a) 3 cm (b) 4 cm (c) 6 cm (d) 8 cm
3. Which one is NOT a valid test for similar triangles? (a) AA (b) SSS (c) SAS (d) SSA
4. AD bisects angle A, with AB = 6 cm, AC = 4 cm and BC = 10 cm. Then BD equals: (a) 4 cm (b) 5 cm (c) 6 cm (d) 7 cm
5. Two similar triangles have corresponding sides in the ratio 3 : 5. The ratio of their areas is: (a) 3 : 5 (b) 9 : 25 (c) 25 : 9 (d) 6 : 10
6. A line cuts sides AB and AC so that AD/DB = AE/EC. The line is: (a) perpendicular to BC (b) parallel to BC (c) a bisector of BC (d) equal to BC

Short and long questions

1. The angles of a triangle are in the ratio 1 : 2 : 6. Find all three angles. (Book, Exercise 14.1 Q1)
2. In triangle ABC, DE || BC, AD = 1.5 cm, BD = 3 cm, AE = 1.3 cm. Find CE. (Book, Exercise 14.1 Q5)
3. AD bisects angle A of triangle ABC, AB = 9 cm, AC = 10 cm, BC = 12 cm. Find BD and DC. (Book, Exercise 14.1 Q7)
4. Triangle ABC ~ triangle DEF, AC = 12 cm, DF = 9 cm, AB = 4 cm, BC = 10 cm. Find DE and EF. (Book, Exercise 14.1 Q12)
5. In triangle RST, KL || ST with RK = 5 cm, KS = 10 cm, RT = 18 cm. Find RL. (Book, Exercise 14.1 Q14)
6. Prove Theorem 14.1, and state its converse. (Long question)

Answer key

MCQs: 1(b), 2(c) - 2/4 = 3/EC gives EC = 6, 3(d), 4(c) - BD/DC = 6/4 with BD + DC = 10 gives BD = 6, 5(b), 6(b).

Questions: 1) 20, 40 and 120 degrees. 2) CE = 2.6 cm. 3) BD = 108/19 = 5.68 cm, DC = 120/19 = 6.32 cm. 4) DE = 3 cm, EF = 7.5 cm. 5) RK/RS = RL/RT gives 5/15 = RL/18, so RL = 6 cm. 6) See pages 4-5 of these notes.

> **Final tip:** For every theorem question in the paper, write Given, To Prove, Construction (if any), then Statements and Reasons in two columns - the book's own format. Most of the marks are for these steps, not for the last line.